

1.
如图 2-1 所示电路, 计算电压 u 。

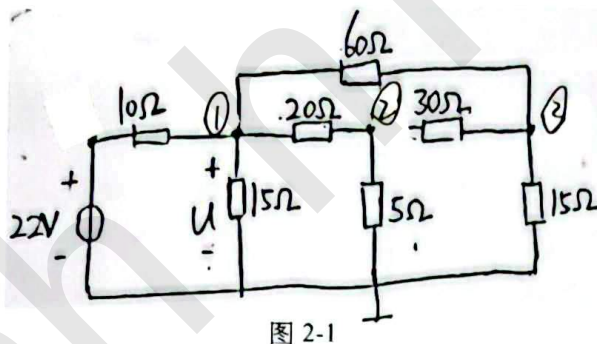


图 2-1

选取节点如图, 可列

$$\begin{cases} (\frac{1}{10} + \frac{1}{15} + \frac{1}{20} + \frac{1}{60})U_{n1} - \frac{1}{20}U_{n2} - \frac{1}{60}U_{n3} = \frac{22}{10} \\ -\frac{1}{20}U_{n1} + (\frac{1}{20} + \frac{1}{30} + \frac{1}{5})U_{n2} - \frac{1}{30}U_{n3} = 0 \\ -\frac{1}{60}U_{n1} - \frac{1}{30}U_{n2} + (\frac{1}{60} + \frac{1}{30} + \frac{1}{15})U_{n3} = 0 \end{cases} \Rightarrow \begin{cases} U_{n1} = 10V \\ U_{n2} = 2V \\ U_{n3} = 2V \end{cases}$$

$$u = U_{n1} = 10V$$

2.
如图 2-2 所示电路, 计算: (1) 节点电压 u_{n1} , u_{n2} , u_{n3} ; (2) 计算 16V 独立电压源提供的功率。

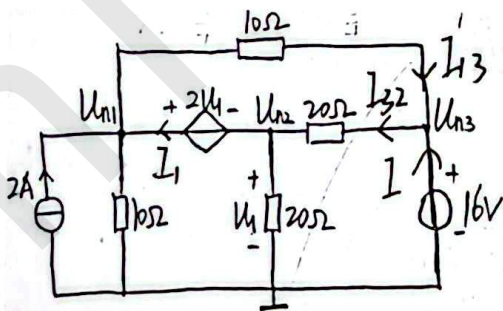


图 2-2

设受控源流过电流为 I_1 , 可列节点电压

$$(1) \begin{cases} (\frac{1}{10} + \frac{1}{10})U_{n1} - \frac{1}{10}U_{n3} = 2 + I_1 \\ (\frac{1}{20} + \frac{1}{20})U_{n2} - \frac{1}{20}U_{n3} = -I_1 \\ U_{n3} = 16 \end{cases}$$

$$\text{补充} \begin{cases} U_{n1} - U_{n2} = 2U_1 \\ U_{n2} = U_1 \end{cases}$$

$$\text{有 } U_{n1} = 18.86V, U_{n2} = 6.29V, U_{n3} = 16V, I_1 = 0.174A$$

设流过 16V 电压源为 I

$$(2) \cdot I_{32} = \frac{U_{n3} - U_{n2}}{20} = 0.4855A$$

$$I_{13} = \frac{U_{n1} - U_{n3}}{10} = 0.286A$$

$$I + I_{13} = I_{32} \Rightarrow I = 0.1995A$$

$$P = U_{n3} \cdot I = 3.192W, \text{发出} \dots$$

3.

试求如图 2-3 所示电路中的电流 I

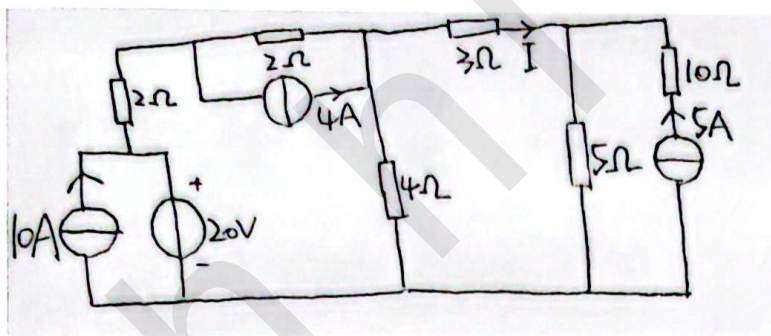
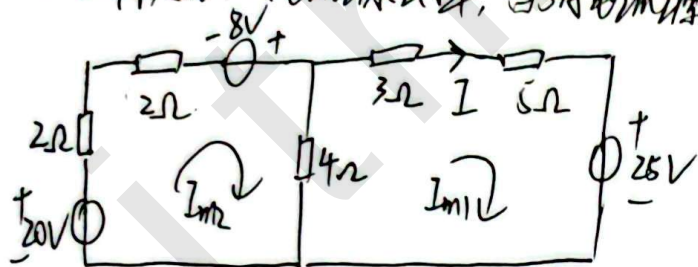


图 2-3

与 20V 并联的 10A 电流源去掉，与 5A 电流源串联的电阻去掉，并将两个诺顿化为戴维南



$$\begin{cases} 12I_{m1} - 4I_{m2} = -25 \\ -4I_{m1} + 8I_{m2} = 28 \end{cases} \Rightarrow \begin{cases} I_{m1} = -1.1A \\ I_{m2} = 2.95A \end{cases}$$

$$I = -1.1A$$

4.

求图 2-4 所示电路中的电流 I_1, I_2, I_3 和 I_4

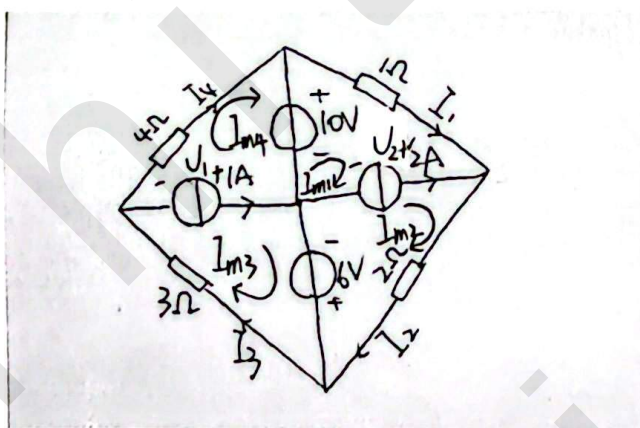


图 2-4

选取回路电流方向如图，设电压源电压为 V_1, V_2 ，可列

$$\begin{cases} I_{m1} = 10 - V_2 \\ 2I_{m2} = V_2 - 6 \\ 3I_{m3} = V_1 + 6 \\ 4I_{m4} = -V_1 - 10 \end{cases}$$

补充

$$\begin{cases} I_{m2} - I_{m1} = 2 \\ I_{m3} - I_{m4} = 1 \end{cases}$$

$$\Rightarrow \begin{cases} I_{m1} = 0 \\ I_{m2} = 2A \\ I_{m3} = 0 \\ I_{m4} = -1A \end{cases} \Rightarrow \begin{cases} I_1 = 0 \\ I_2 = 2A \\ I_3 = 0 \\ I_4 = -1A \end{cases}$$